

ARCHIT SHARMA

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EDUCATION

New York University

Sept 2024 - May 2026

M.S. Mechatronics and Robotics | Advisor: Joo H. Kim

GPA: 3.8

Birla Institute of Technology Pilani - Goa

Aug 2019 - May 2023

B.Tech Mechanical Engineering

GPA: 3.4

TECHNICAL SKILLS

Robotics: ROS/ROS2, NVIDIA Isaac Sim/Isaac Lab, MuJoCo, VLAs, Reinforcement Learning

Software: Python, C/C++, Cloud & HPC

ML & AI: PyTorch, JAX, LLMs, Computer Vision & Perception

EXPERIENCE

Graduate Researcher, AI4CE Lab

July 2026 - Present

- Developed an in-context learning framework for VLAs that retrieves context chunks by task progress, raising out-of-domain task success from 2.7% to 25.0%.
- Designed and trained an in-context value function estimator that scores task progress using a reference demonstration, achieving 17% higher correlation with human labels than the best existing benchmark.
- Built a distributed deployment pipeline for the YOR bimanual robot, splitting inference, vision, and motion control, and collected a 2.6M frame, 36-task teleoperated dataset for training and validation.

Graduate Researcher, Applied Dynamics and Optimization Laboratory

Sept 2025 - July 2026

- Built a pipeline coupling COMSOL multiphysics simulations with MuJoCo's rigid-body dynamics for a bio-inspired fish robot, enabling a parameter sweep that identified a gait with 22% lower cost of transport.
- Implemented a convex optimization framework that projects the fish's CFD-derived link loads onto a dynamically feasible set, minimizing the deviation to within 5% of the original.
- Generated reference walking trajectories for a Unitree G1 humanoid, using Mink, as imitation targets for an RL policy, tracking CoM and feet objectives within 1 cm RMS error.

Undergraduate Researcher, Dynamics and Vibrations Lab

Aug 2021 - Jan 2024

- Designed a self-learning active suspension controller using an adaptive neural network trained via leaky average gradient descent and replay buffer, reducing peak force by 63% against the strongest baseline.
- Built a custom MATLAB-Simulink framework for controller testing, benchmarking the neural-network controller against 5 classical baselines (PD, PID, LQR, sliding-mode, sky-hook), across 5 road profiles.

PUBLICATIONS

[ICRA'27] L Macesanu, J Zhang, T Zhang, A Patil, K Angers, **A Sharma**, K Imam, G Perez-Haiek, C Clark, C Feng. "VICTR: Value-Informed In-Context Learning for Vision-Language-Action Models," IEEE International Conference on Robotics and Automation, under review.

[ICSV'24] **A Sharma**, P M Singru. "Neural Network-Based Control for Vehicle Suspension Systems with Active Damping," International Congress on Sound and Vibration.

SELECTED PROJECTS

Quadruped RL Policy for Locomotion

Sept 2025 - Aug 2026

- Trained a PPO locomotion policy for the Unitree Go2 on rough and stepped terrain with custom Raibert-heuristic, feet-clearance, and contact-force rewards, transferring it to hardware via domain randomization and a PD actuator model.

Vision-Based Maze Navigation Agent

Sept 2025 - Dec 2025

- Built a visual maze-navigation pipeline, precomputing a kNN graph over ViT embeddings of exploration images, and training a DQN agent to traverse the environment using the graph representation, reaching the target in over 50% of runs.